DATE: _____

EX: 9 THE BOX–JENKINS MODEL (ARIMA)

AIM

To understand and implement the Box–Jenkins approach to time-series forecasting using an ARIMA model.

ALGORITHM

1. Import the required libraries.

2. Create monthly sales data and plot the time series.

3. Check stationarity using the ADF test.

4. Apply first differencing and test stationarity again.

5. Plot ACF and PACF.

6. Fit an ARIMA(1,1,1) model.

7. Display AIC and BIC values.

8. Check residuals and perform the Ljung–Box test.

9. Forecast the next six months and plot the forecast.

PROGRAM

```
import warnings
warnings.filterwarnings("ignore")
import pandas as pd
import matplotlib.pyplot as plt
from statsmodels.tsa.arima.model import ARIMA
from statsmodels.tsa.stattools import adfuller
from statsmodels.graphics.tsaplots import plot_acf, plot_pacf
from statsmodels.stats.diagnostic import acorr_ljungbox

sales = [100,105,103,110,115,112,120,125,123,130,135,132,
         140,145,143,150,155,152,160,165,163,170,175,172]
data = pd.Series(sales, name="Sales")

plt.figure(figsize=(10,4))
plt.plot(data, marker="o")
plt.title("Monthly Sales Data")
plt.xlabel("Month"); plt.ylabel("Sales")
plt.grid(True); plt.tight_layout(); plt.show()

adf_result = adfuller(data)
print("ADF Statistic:", round(adf_result[0], 4))
print("p-value:", round(adf_result[1], 4))

diff_data = data.diff().dropna()
print(diff_data.to_string(index=False))
adf_diff = adfuller(diff_data)
print("Differenced ADF Statistic:", round(adf_diff[0], 4))
print("Differenced p-value:", round(adf_diff[1], 4))

plot_acf(diff_data, lags=10); plt.show()
plot_pacf(diff_data, lags=10, method="ywm"); plt.show()

model = ARIMA(data, order=(1,1,1))
fitted_model = model.fit()
print(fitted_model.summary())
print("AIC:", round(fitted_model.aic, 4))
print("BIC:", round(fitted_model.bic, 4))

residuals = fitted_model.resid
plt.plot(residuals)
plt.axhline(0, color="red", linestyle="--")
plt.title("ARIMA Model Residuals")
plt.grid(True); plt.show()
plot_acf(residuals, lags=10); plt.show()
print(acorr_ljungbox(residuals, lags=[5], return_df=True))

forecast = fitted_model.get_forecast(steps=6)
forecast_mean = forecast.predicted_mean
forecast_ci = forecast.conf_int()
for month, value in enumerate(forecast_mean, start=25):
    print(f"Month {month}: {value:.2f}")
print(forecast_ci.round(2))

future_months = list(range(len(data), len(data)+6))
plt.plot(range(len(data)), data.values, marker="o", label="Actual Sales")
plt.plot(future_months, forecast_mean.to_numpy(), marker="o",
         linestyle="--", label="Forecast")
plt.fill_between(future_months, forecast_ci.iloc[:,0].to_numpy(),
                 forecast_ci.iloc[:,1].to_numpy(), alpha=0.2,
                 label="95% Confidence Interval")
plt.title("Box-Jenkins ARIMA: Six-Month Sales Forecast")
plt.xlabel("Month"); plt.ylabel("Sales")
plt.legend(); plt.grid(True); plt.show()
```

DATE: _____

OUTPUT

ADF Test on Original Data

ADF Statistic: -0.3013
p-value: 0.9253

First-Differenced Data

5.0, -2.0, 7.0, 5.0, -3.0, 8.0, 5.0, -2.0, 7.0, 5.0, -3.0, 8.0, 5.0, -2.0, 7.0, 5.0, -3.0, 8.0, 5.0, -2.0, 7.0, 5.0, -3.0

ADF Test After Differencing

ADF Statistic: -47375573695773.04
p-value: 0.0

Model Selection Criteria

Model: ARIMA(1,1,1)
AIC: 142.4652
BIC: 145.8717

Ljung–Box Test

Test statistic (lag 5): 0.555286
p-value: 0.989959

Forecast for the Next Six Months

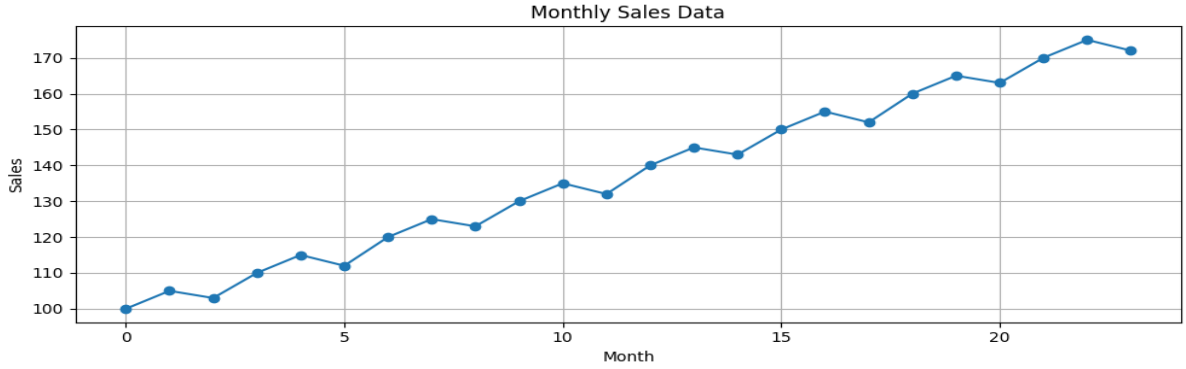
Month	Forecast	Lower 95% CI	Upper 95% CI
25	170.05	161.11	178.99
26	170.94	154.78	187.10
27	170.54	150.76	190.31
28	170.72	147.41	194.03
29	170.64	144.46	196.81
30	170.67	141.83	199.52

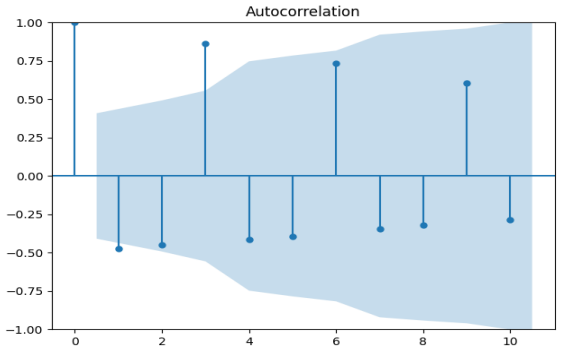
Model Parameters

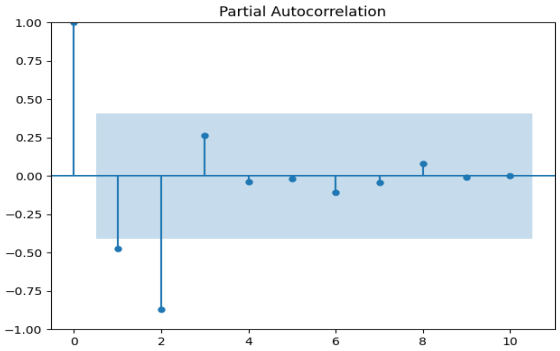
ar.L1 = -0.4562; ma.L1 = 0.9996; sigma^2 = 20.0362

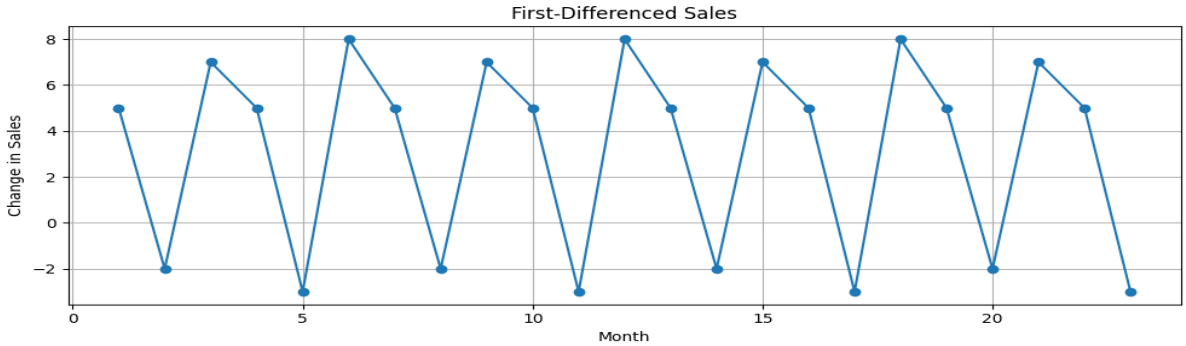
DATE: _____

GRAPHS


Monthly Sales Data
Sales
Month
100
110
120
130
140
150
160
170
0
5
10
15
20

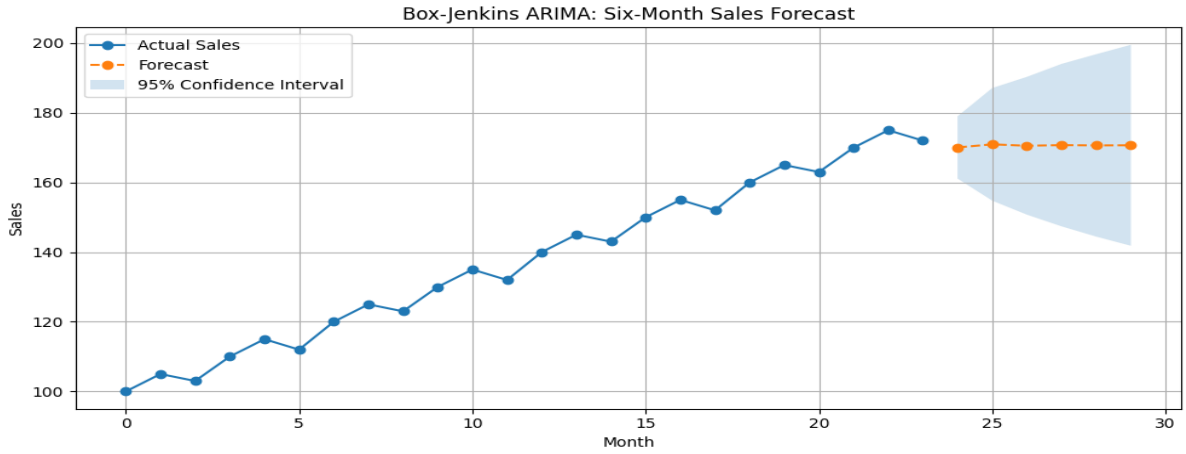

Autocorrelation
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
0
2
4
6
8
10


Partial Autocorrelation
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
0
2
4
6
8
10


First-Differenced Sales
Change in Sales
Month
8
6
4
2
0
−2
0
5
10
15
20

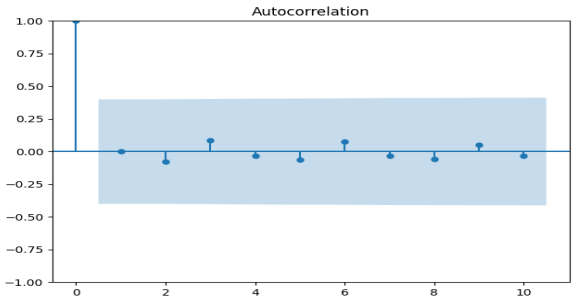
DATE: _____

GRAPH



RESIDUAL GRAPHS





OBSERVATION

The original sales series shows an upward trend. The ADF test gives a p-value of 0.9253, indicating that the original series is non-stationary. After first differencing, the ADF p-value is 0.0. The ARIMA(1,1,1) model was fitted and used to forecast the next six months. The Ljung–Box test at lag 5 gives a p-value of 0.989959.

RESULT

Thus, the Box–Jenkins ARIMA model was implemented, checked, and used to forecast six future sales values.